Project: Worked examples
Title: Snap-fits
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## 1. Scope and method

Calculates the strain, deflection force and joining forces of a plastic snap-fit: a cantilever lug, an annular ring or a torsion bar. It checks the strain against the permissible strain and whether the joint is self-locking.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Cantilever geometry (Euler–Bernoulli) | $y = K_y \cdot \varepsilon \cdot l^2/h$ with $K_y = 2/3$ (constant section; the design guides print the rounded 0.67); a linear taper of the thickness to $r_h = h_{tip}/h$ gives $K_y = 2 \cdot J/\varphi_{max}$ with $J = [2r_h - r_h^2/2 - \ln r_h - 3/2]/(1 - r_h)^3$ and $\varphi_{max} = 1$ for $r_h \geq 1/2$ (otherwise the strain peaks inside the beam at $t = (2a - 1)/a$, $a = 1 - r_h$, with $\varphi_{max} = (1 - t)/(1 - a \cdot t)^2$); a linear taper of the width to $r_b$ gives $K_y = 2$... |
| Strain and force of the cantilever | $\varepsilon = y \cdot h/(K_y \cdot l^2)$; $F_{defl} = b \cdot h^2 \cdot Es \cdot \varepsilon/(6 \cdot l \cdot \varphi_{max}) = (b \cdot h^2/6) \cdot (Es \cdot \varepsilon/l)$ at the root of a constant section; Es is the secant modulus at ε (typed, or interpolated from the W3 points Es(ε)) |
| Annular (derived, free thin ring) | nominal hoop strain $\varepsilon_{nom} = \Delta D/D$ (the mean diameter D widened by the diametral interference ΔD); the strain checked is the larger of $\varepsilon = K_{ann} \cdot \varepsilon_{nom}$ ($K_{ann}$ the user's strain factor) and the inner-surface strain $\Delta D/(D - h)$; the radial contact force summed round the ring is $F_{defl} = K_F \cdot 2\pi \cdot Es \cdot \varepsilon_{nom} \cdot h \cdot b$ (hoop force $Es \cdot \varepsilon_{nom} \cdot h \cdot b$ over the radius, $K_F$ the user's optional force factor, 1 if none), b bein... |
| Torsion bar of radius r with a lever of length a deflected y | $\varphi = y/a$; the bar is held at one end (free length $l_{eff} = l$) or at both ends with the lever in the middle ($l_{eff} = l/2$, two halves in parallel); $\gamma = \varphi \cdot r/l_{eff}$, equivalent tensile strain $\varepsilon = \gamma/(1 + \nu)$, $\gamma_{perm} = \varepsilon_{perm} \cdot (1 + \nu)$; $G = Es/(2(1 + \nu))$, $T = n \cdot G \cdot (\pi r^4/2) \cdot \varphi/l_{eff}$ with n = 1 or 2, $F_{defl} = T/a$ (both ends: $F_{defl} = 4 \cdot G \cdot J \cdot \varphi/(l \cdot a)$) |
| Joining forces | $W = F_{defl} \cdot (\mu + \tan \alpha)/(1 - \mu \cdot \tan \alpha)$ on the entry face (angle α to the joining direction) and $W' = F_{defl} \cdot (\mu + \tan \alpha')/(1 - \mu \cdot \tan \alpha')$ on the return face (α′); the joint is self-locking, with an infinite separation force, when $\alpha' \geq \arctan(1/\mu)$; assembly jams when $\mu \cdot \tan \alpha \geq 1$ |
| Utilisation and permissible deflection | $u = \varepsilon/\varepsilon_{perm}$ (FAIL above 1); $y_{perm} = K_y \cdot \varepsilon_{perm} \cdot l^2/h$ (annular $\Delta D_{perm} = \varepsilon_{perm} \cdot \min(D/K_{ann}, D - h)$; torsion $y_{perm} = \gamma_{perm} \cdot l_{eff} \cdot a/r$) |

Basis: Cantilever, annular and torsion snap-fit relations[1](Ch. B).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| mode | Form of snap-fit | Cantilever, tapered thickness | | drawing: ENC-310-07 rev C | any |
| l | Length of the beam (cantilever); of the torsion bar: its free length when held at one end, or its whole length between the two supports when held at both ends with the lever in the middle | 18 | mm | drawing: ENC-310-07 rev C | 2 … 100 mm |
| h | Thickness at the root (cantilever); wall thickness of the ring (annular) | 2.2 | mm | drawing: ENC-310-07 rev C | 0.3 … 10 mm |
| b | Width at the root (cantilever); effective axial length of the ring (annular: that of a free ring, not the lip of a longer tube) | 7 | mm | drawing: ENC-310-07 rev C | 1 … 50 mm |
| y | Deflection at the tip (cantilever) or at the end of the lever (torsion): the undercut to be cleared | 1.2 | mm | drawing: ENC-310-07 rev C | 0.05 … 20 mm |
| $h_{ratio}$ | Tip thickness / root thickness (tapered thickness) | 0.5 | | drawing: ENC-310-07 rev C | 0.25 … 1 |
| $Es_{source}$ | Where the secant modulus comes from: typed at the design strain, or interpolated from the Es(ε) points of a material record | Typed secant modulus | | assumption: Secant modulus typed for this example; a W3 material with its Es(ε) points can be applied instead. | any |
| loading | How often the joint is made: selects the one-time or the repeated permissible strain | One-time (assembled once or rarely) | | spec: Enclosure ENC-310 §6: the lid is opened for service only | any |
| $T_{use}$ | Temperature of use. Recorded: Es and the permissible strain must be the values at this temperature | 23 | °C | spec: Enclosure ENC-310 §6: room temperature | -40 … 120 °C |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| Es | Secant modulus at the design strain and temperature (annular: at the nominal hoop strain ΔD/D) (typed; no default: the plastic's stress–strain data) | 2000 | MPa | assumption: Example secant modulus at the design strain and temperature. Enter Es for your plastic from its stress–strain data or a W3 material; no plastics data are shipped. | 10 … 20000 MPa |
| $eps_{perm,once}$ | Permissible strain for one-time joining, as a fraction, for example 0.04, at the temperature of use (no default: from the plastic's data) | 0.04 | | assumption: Example permissible strain for one-time joining. Enter it for your plastic at the temperature of use from its data or a W3 material; no plastics data are shipped. | 0.001 … 0.3 |
| α | Entry (assembly) angle α of the inclined face to the joining direction | 30 | ° | drawing: ENC-310-07 rev C | 5 … 60 ° |
| $\alpha_r$ | Return (separation) angle α′ of the retaining face (90° is a face square to the joining direction) | 65 | ° | drawing: ENC-310-07 rev C | 10 … 90 ° |
| μ | Coefficient of friction between the two parts (record snapfit.friction; no default: the pair of materials, from your reference or a W3 friction pair) | 0.35 | | assumption: Example friction coefficient for a plastic-on-plastic pair. Enter μ for your pair of materials and surfaces from your reference or a test (record snapfit.friction). | 0.05 … 1 |

## 3. Assumptions

A1. The cantilever follows Euler–Bernoulli beam theory, with a rigid root, a linear taper and small deflection. (Euler–Bernoulli beam theory (derived); the guide's closed forms for the constant section are to be checked against the owner's copy.)
A2. The secant modulus is constant over the whole beam, at the design strain. (One modulus stands for the stress–strain curve.)
A3. The strain is the nominal surface strain at the critical section, with no local concentration from the root fillet. (The geometry factors are derived by integration, and the 0.67 of the design guides is the constant-section 2/3 rounded.)
A4. Es and the permissible strain are the plastic's values for the temperature, loading and moisture of use. (No plastics data are shipped, so they come from the plastic maker's data or a material record.)
A5. The annular snap is a free, thin, closed ring of effective axial length b. (Derived from ring geometry, not a relation of the design guide, so compare it with the owner's copy of the guide.)

A6. The nominal hoop strain is the diametral interference over the mean diameter, ΔD/D. (Derived from ring geometry.)
A7. The strain checked is the larger of $K_{ann}$ times the nominal strain and the inner-surface strain ΔD/(D − h). ($K_{ann}$ is the user's strain factor (record snapfit.annular-factor, no default), and the inner-surface strain is slightly above the mean-diameter value.)
A8. The contact force is the hoop force $Es·\varepsilon_{nom}·h·b$ taken round the ring, times the optional force factor $K_F$. ($K_F$ is the user's record (blank is 1), and a tube behind the ring stiffens the joint, which only $K_F$ can represent.)
A9. The torsion bar is solid and round, loaded at a lever end, with a small angle φ = y/a. (γ = φ·r/l and $\gamma_{perm} \approx \varepsilon_{perm}·(1 + \nu)$ are the brief's design-guide relations, to be checked against the owner's copy.)
A10. The bar is held at one end, or at both ends with the lever in the middle. (Held at both ends it has twice the strain and four times the stiffness of one end for the same total length, and the two-ended case is derived.)
A11. The permissible shear strain is $\varepsilon_{perm}·(1 + \nu)$, and the equivalent tensile strain is γ/(1 + ν). (G = Es/(2(1 + ν)) uses the secant modulus at that strain.)
A12. The mating part is rigid, with Coulomb friction μ on the inclined face (α for assembly, α′ for separation). (Friction on an inclined plane, with the friction coefficient the user's record.)
A13. The deflection force is taken at the design deflection, with no creep, stress relaxation or impact effect in service. (A static joint model.)
A14. $Es_{source}$ (Where the secant modulus comes from: typed at the design strain, or interpolated from the Es(ε) points of a material record) is taken as Typed secant modulus. (Secant modulus typed for this example; a W3 material with its Es(ε) points can be applied instead.)
A15. Es (Secant modulus at the design strain and temperature (annular: at the nominal hoop strain ΔD/D) (typed; no default: the plastic's stress–strain data)) is taken as 2000 MPa. (Example secant modulus at the design strain and temperature. Enter Es for your plastic from its stress–strain data or a W3 material; no plastics data are shipped.)
A16. $eps_{perm,once}$ (Permissible strain for one-time joining, as a fraction, for example 0.04, at the temperature of use (no default: from the plastic's data)) is taken as 0.04. (Example permissible strain for one-time joining. Enter it for your plastic at the temperature of use from its data or a W3 material; no plastics data are shipped.)
A17. μ (Coefficient of friction between the two parts (record snapfit.friction; no default: the pair of materials, from your reference or a W3 friction pair)) is taken as 0.35. (Example friction coefficient for a plastic-on-plastic pair. Enter μ for your pair of materials and surfaces from your reference or a test (record snapfit.friction).)
A18. Es and the permissible strain are the plastic's values at 23 °C, the temperature of use. The sheet cannot check them, or the moisture state of the part. (Recorded temperature of use.)

## 4. Working

[10] $l_h$ = l / h
= 18 mm / 2.2 mm
= 8.1818
Slenderness of the beam l/h

[12] J = [2r − r²/2 − ln r − 3/2]/(1 − r)³ ²((derived))
= r = 0.5
= 0.54518
Deflection integral J of the thickness taper (derived: Euler–Bernoulli integration)

[13] $\varphi_{max}$ = (1 − t*)/(1 − α·t*)², t* = (2α − 1)/α for α > 1/2, else 1 ²((derived))
= α = 0.5
= 1
Largest value of the strain shape along the beam (1 at the root)

[14] $K_y$ = 2·J/$\varphi_{max}$ ²((derived))
= 2 × 0.54518 / 1
= 1.0904

Deflection factor: $y = K_y \cdot \varepsilon \cdot l^2 / h$

[15] $eps_{max} = y * h / (K_y * l^2)$ [2]((derived))

$= 1.2\ mm \times 2.2\ mm / (1.0904 \times (18\ mm)^2)$

$= 0.74729\ \%$

Maximum surface strain at the critical section (nominal)

[18] $eps_{perm} = eps_{perm,once}$ [1](Plastics: A Design Guide)

$= 0.04$

$= 4\ \%$

Permissible strain (one-time joining, as entered)

[21] $F_{defl} = b * h^2 * Es * eps_{max} / (6 * l * \varphi_{max})$ [2]((derived))

$= 7\ mm \times (2.2\ mm)^2 \times 2000\ MPa \times 0.74729\ \% / (6 \times 18\ mm \times 1)$

$= 4.6886\ N$

Deflection force at the tip: $F_{defl} = (b \cdot h^2/6) \cdot (Es \cdot \varepsilon / l)$ at the root of a constant section

[22] $y_{perm} = K_y * eps_{perm} * l^2 / h$ [2]((derived))

$= 1.0904 \times 4\ \% \times (18\ mm)^2 / 2.2\ mm$

$= 6.4232\ mm$

Permissible tip deflection

[23] $strain_{util} = eps_{max} / eps_{perm}$

$= 0.74729\ \% / 4\ \%$

$= 0.18682$

Strain / permissible strain

[24] $tan_{\alpha} = tan(\alpha)$

$= tan(30\ deg)$

$= 0.57735$

tan α of the entry face

[25] $\mu_{tan,\alpha} = \mu * tan_{\alpha}$

$= 0.35 \times 0.57735$

$= 0.20207$

μ·tan α of the entry face

[26] $assembly_{force} = F_{defl} * (\mu + tan_{\alpha}) / (1 - \mu_{tan,\alpha})$ [1](Plastics: A Design Guide)

$= 4.6886\ N \times (0.35 + 0.57735) / (1 - 0.20207)$

$= 5.4491\ N$

Assembly force $W = F_{defl} \cdot (\mu + \tan\alpha)/(1 - \mu \cdot \tan\alpha)$

[27] $\alpha_{lock} = arctan(1/\mu)$ [1](Plastics: A Design Guide)

$= arctan(1/0.35)$

$= 70.71\ deg$

Return angle above which the joint is self-locking (π/2 when μ = 0)

[28] $tan_{ar} = tan(\alpha_r)$

$= tan(65\ deg)$

$= 2.1445$

tan α′ of the return face

[29] $retention_{force} = F_{defl} * (\mu + tan_{ar}) / (1 - \mu * tan_{ar})$ [1](Plastics: A Design Guide)

$= 4.6886\ N \times (0.35 + 2.1445) / (1 - 0.35 \times 2.1445)$

$= 46.891\ N$

Separation force $W' = F_{defl} \cdot (\mu + \tan\alpha')/(1 - \mu \cdot \tan\alpha')$

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $assembly_{force}$ | 5.449 N | Assembly force |
| $retention_{force}$ | 46.89 N | Separation force |
| $strain_{util}$ | 0.19 | Strain / permissible strain |
| $eps_{max}$ | 0.7473 % | Maximum strain (governing) |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $eps_{perm}$ | 4 % | Permissible strain |
| Es | 2000 MPa | Secant modulus used |
| $F_{defl}$ | 4.689 N | Deflection force |
| $y_{perm}$ | 6.423 mm | Permissible deflection |
| $l_h$ | 8.1818 | Slenderness l/h |
| $K_y$ | 1.0904 | Deflection factor of the cantilever |
| $\varphi_{max}$ | 1 | Strain shape maximum |
| $\alpha_{lock}$ | 70.71° | Self-locking return angle |
| $self_{locking}$ | "No" | Self-locking |
| $fillet_{advisory}$ | "The strain above is the nominal strain, with no local concentration. A sharp internal corner at the root (or at the end of the bar or ring) concentrates the strain and is the usual place for a snap-fit to break: give the root a generous fillet radius, of the order of the thickness, taper the section away from it, and apply a concentration factor from your own reference if the radius is small." | Root-fillet advisory |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Strain at the critical section against the permissible strain | 0.7473 % | 4 % | 0.19 | [1](Plastics: A Design Guide) |
| PASS | Entry-face friction μ·tan α against the jamming limit | μ·tan α = 0.20207 | < 1 | – | [1](Plastics: A Design Guide) |

## 7. Applicability

WARNING records: tabulated values not taken from a reference record are $eps_{perm,once}$ (Material properties); μ (Coefficient of friction); Es (Secant modulus at the design strain). Enter each as one so its source, edition and licence travel with it. [record]

## 8. Verification

```
Template mech.snap-fit@1.5.0 Content hash 32eaeb1d9537950d Engine calc-engine/1.0.0
Known-answer cases passed 11/11 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. Covestro, Snap-Fit Joints for Plastics: A Design Guide.
2. Euler–Bernoulli beam theory.